CHAPTER 4

Color Scales

There are three fundamental use cases for color in data visualizations: we can use color to distinguish groups of data from each other, to represent data values, and to highlight. The types of colors we use and the way in which we use them are quite different for these three cases.

Color as a Tool to Distinguish

We frequently use color as a means to distinguish discrete items or groups that do not have an intrinsic order, such as different countries on a map or different manufacturers of a certain product. In this case, we use a *qualitative* color scale. Such a scale contains a finite set of specific colors that are chosen to look clearly distinct from each other while also being equivalent to each other. The second condition requires that no one color should stand out relative to the others. Also, the colors should not create the impression of an order, as would be the case with a sequence of colors that get successively lighter. Such colors would create an apparent order among the items being colored, which by definition have no order.

Many appropriate qualitative color scales are readily available. Figure 4-1 shows three representative examples. In particular, the ColorBrewer project provides a nice selection of qualitative color scales, including both fairly light and fairly dark colors [Brewer 2017].

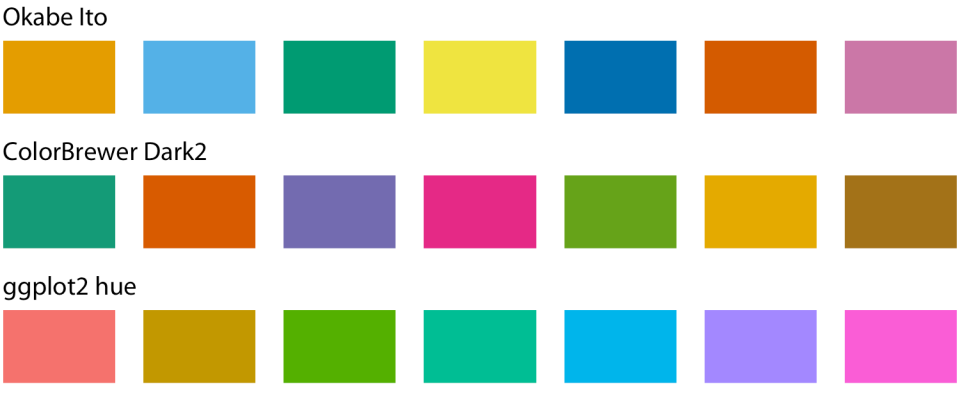


Figure 4-1. Example qualitative color scales. The Okabe Ito scale is the default scale used throughout this book [Okabe and Ito 2008]. The ColorBrewer Dark2 scale is provided by the ColorBrewer project [Brewer 2017]. The ggplot2 hue scale is the default qualitative scale in the widely used plotting software ggplot2.

As an example of how we use qualitative color scales, consider Figure 4-2. It shows the percent population growth from 2000 to 2010 in US states. I have arranged the states in order of their population growth, and I have colored them by geographic region. This coloring highlights that states in the same regions have experienced similar population growth. In particular, states in the West and the South have seen the largest population increases, whereas states in the Midwest and the Northeast have grown much less.

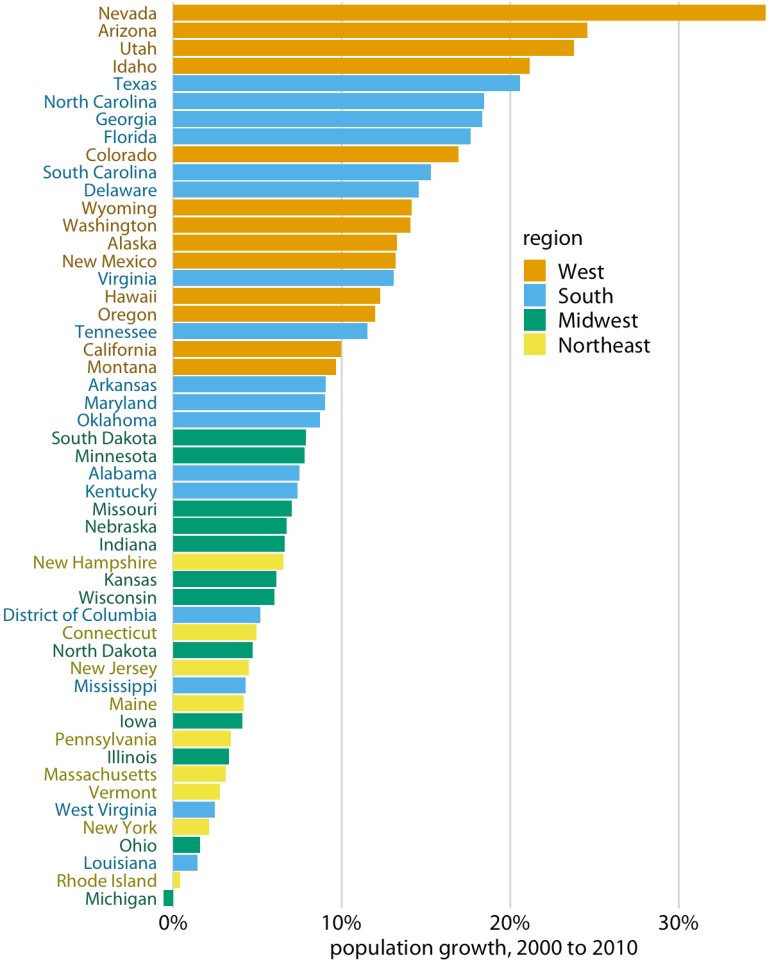


Figure 4-2. Population growth in the US from 2000 to 2010. States in the West and South have seen the largest increases, whereas states in the Midwest and Northeast have seen much smaller increases (or even, in the case of Michigan, a decrease). Data source: US Census Bureau.

Color to Represent Data Values

Color can also be used to represent quantitative data values, such as income, temperature, or speed. In this case, we use a *sequential* color scale. Such a scale contains a sequence of colors that clearly indicate which values are larger or smaller than which other ones, and how distant two specific values are from each other. The second point

implies that the color scale needs to be perceived to vary uniformly across its entire range.

Sequential scales can be based on a single hue (e.g., from dark blue to light blue) or on multiple hues (e.g., from dark red to light yellow) (Figure 4-3). Multihue scales tend to follow color gradients that can be seen in the natural world, such as dark red, green, or blue to light yellow, or dark purple to light green. The reverse (e.g., dark yellow to light blue) looks unnatural and doesn't make a useful sequential scale.



Figure 4-3. Example sequential color scales. The ColorBrewer Blues scale is a monochromatic scale that varies from dark to light blue. The Heat and Viridis scales are multihue scales that vary from dark red to light yellow and from dark blue via green to light yellow, respectively.

Representing data values as colors is particularly useful when we want to show how the data values vary across geographic regions. In this case, we can draw a map of the geographic regions and color them by the data values. Such maps are called *choropleths*. Figure 4-4 shows an example where I have mapped annual median income within each county in Texas onto a map of those counties.

In some cases, we need to visualize the deviation of data values in one of two directions relative to a neutral midpoint. One straightforward example is a dataset containing both positive and negative numbers. We may want to show those with different colors, so that it is immediately obvious whether a value is positive or negative as well as how far in either direction it deviates from zero. The appropriate color scale in this situation is a *diverging* color scale. We can think of a diverging scale as two sequential scales stitched together at a common midpoint, which usually is represented by a light color (Figure 4-5). Diverging scales need to be balanced, so that the progression from light colors in the center to dark colors on the outside is approximately the same in either direction. Otherwise, the perceived magnitude of a data value would depend on whether it fell above or below the midpoint value.

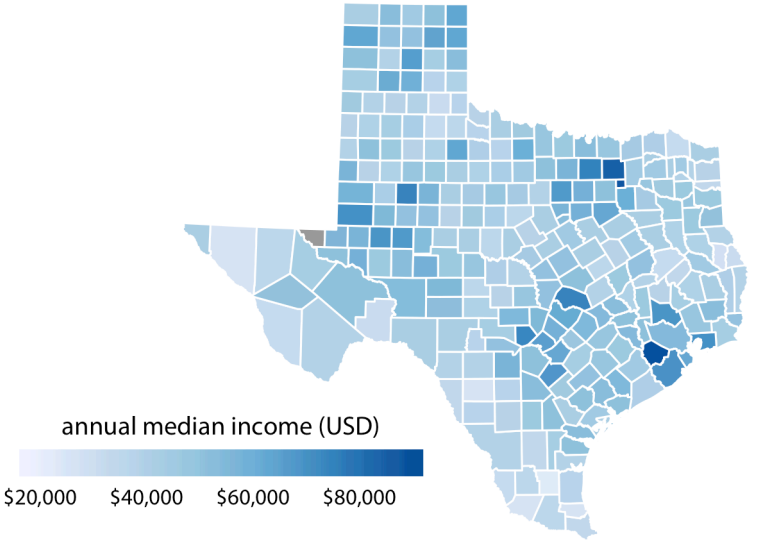


Figure 4-4. Median annual income in Texas counties. The highest median incomes are seen in major Texas metropolitan areas, in particular near Houston and Dallas. No median income estimate is available for Loving County in West Texas, and therefore that county is shown in gray. Data source: 2015 Five-Year American Community Survey.

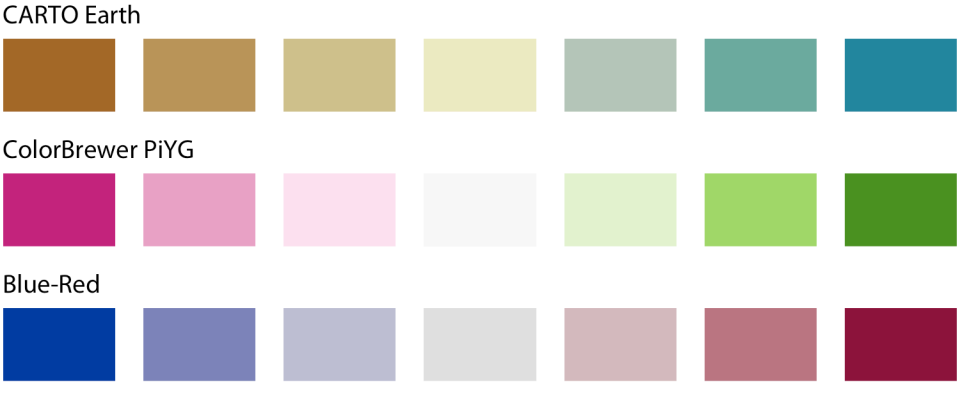


Figure 4-5. Example diverging color scales. Diverging scales can be thought of as two sequential scales stitched together at a common midpoint color. Common color choices for diverging scales include brown to greenish blue, pink to yellow-green, and blue to red.

As an example application of a diverging color scale, consider Figure 4-6, which shows the percentage of people identifying as white in Texas counties. Even though percentage is always a positive number, a diverging scale is justified here, because 50% is a meaningful midpoint value. Numbers above 50% indicate that whites are in the majority and numbers below 50% indicate the opposite. The visualization clearly shows in which counties whites are in the majority, in which they are in the minority, and in which whites and nonwhites occur in approximately equal proportions.

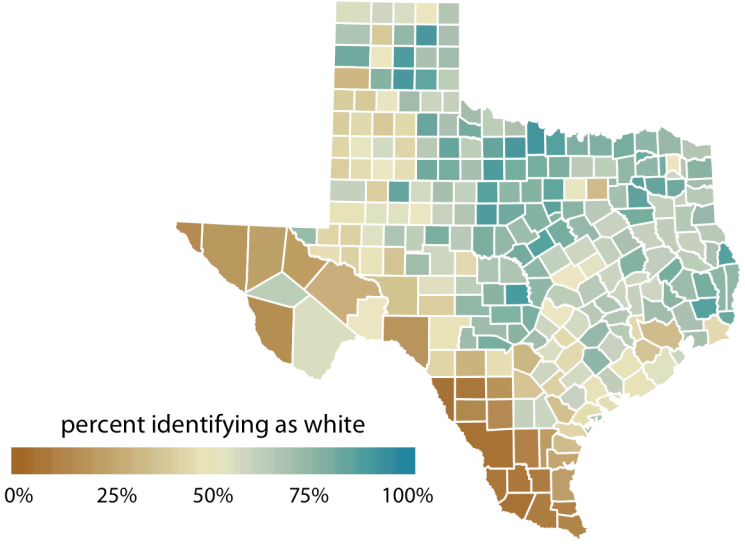


Figure 4-6. Percentage of people identifying as white in Texas counties. Whites are in the majority in North and East Texas but not in South or West Texas. Data source: 2010 US Decennial Census.

Color as a Tool to Highlight

Color can also be an effective tool to highlight specific elements in the data. There may be specific categories or values in the dataset that carry key information about the story we want to tell, and we can strengthen the story by emphasizing the relevant figure elements to the reader. An easy way to achieve this emphasis is to color these figure elements in a color or set of colors that vividly stand out against the rest of the figure. This effect can be achieved with *accent* color scales, which are color scales that contain both a set of subdued colors and a matching set of stronger, darker, and/or more saturated colors (Figure 4-7).

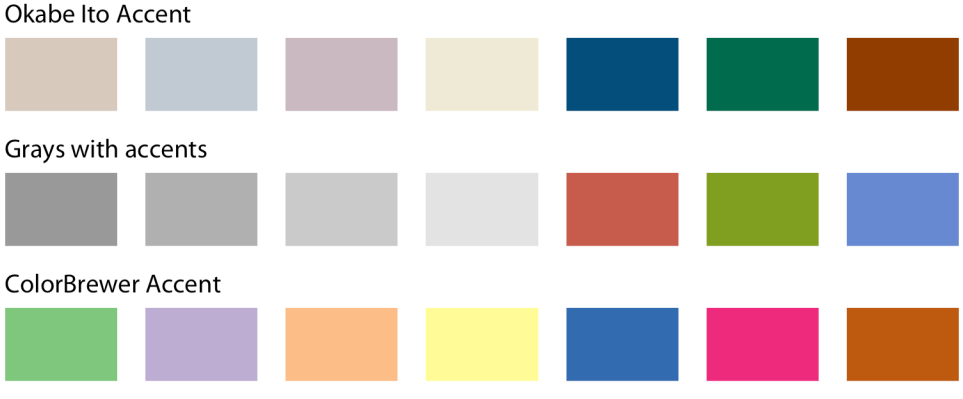


Figure 4-7. Example accent color scales, each with four base colors and three accent colors. Accent color scales can be derived in several different ways: (top) we can take an existing color scale (e.g., the Okabe Ito scale, Figure 4-1) and lighten and/or partially desaturate some colors while darkening others; (middle) we can take gray values and pair them with colors; (bottom) we can use an existing accent color scale (e.g., the one from the ColorBrewer project).

As an example of how the same data can support differing stories with different coloring approaches, I have created a variant of Figure 4-2 where now I highlight two specific states, Texas and Louisiana (Figure 4-8). Both states are in the South, they are immediate neighbors, and yet one state (Texas) was the fifth fastest growing state within the US from 2000 to 2010 whereas the other was the third slowest growing.

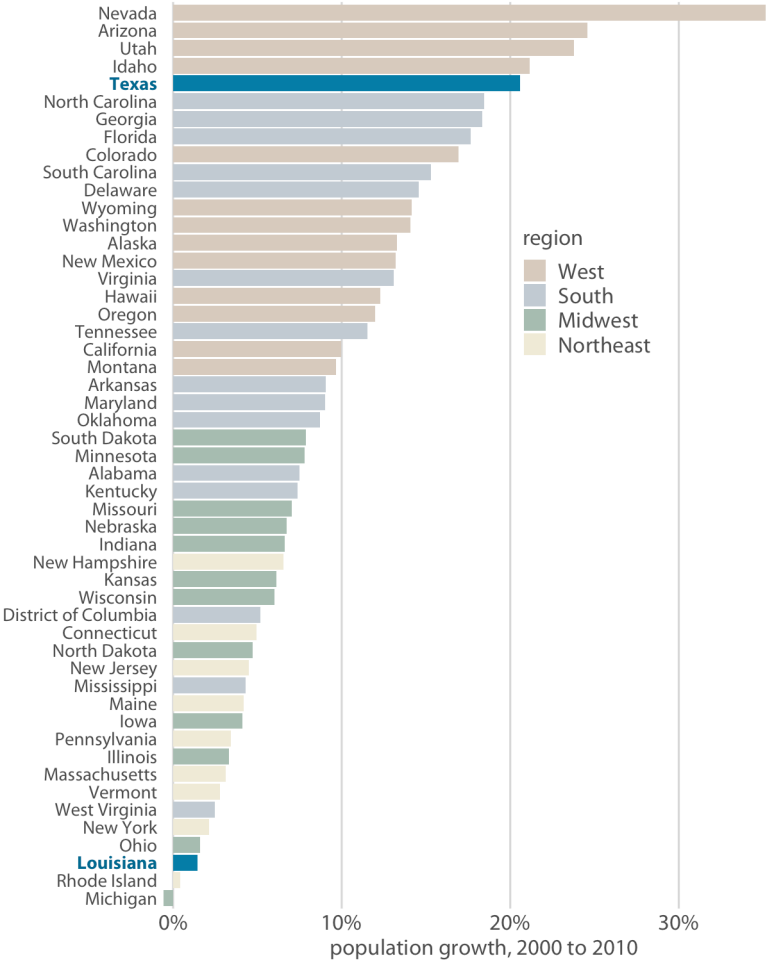


Figure 4-8. From 2000 to 2010, the two neighboring southern states, Texas and Louisiana, experienced among the highest and lowest population growth across the US. Data source: US Census Bureau.

When working with accent colors, it is critical that the baseline colors do not compete for attention. Notice how drab the baseline colors are in Figure 4-8, yet they work well to support the accent color. It is easy to make the mistake of using baseline colors that are too colorful, so that they end up competing for the reader's attention against the accent colors. There is an easy remedy, however: just remove all color from all elements in the figure except the highlighted data categories or points. An example of this strategy is provided in Figure 4-9.

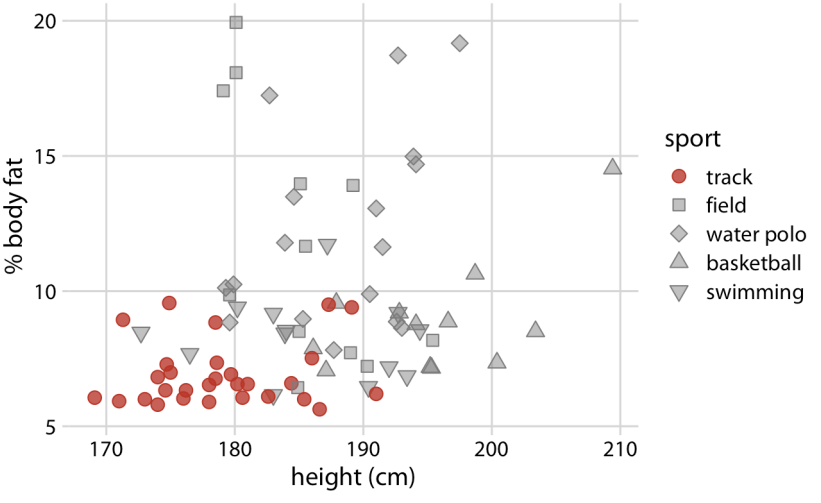


Figure 4-9. Track athletes are among the shortest and leanest of male professional athletes participating in popular sports. Data source: [Telford and Cunningham 1991].

CHAPTER 17

The Principle of Proportional Ink

In many different visualization scenarios, we represent data values by the extent of a graphical element. For example, in a bar plot, we draw bars that begin at 0 and end at the data value they represent. In this case, the data value is not only encoded in the endpoint of the bar but also in the height or length of the bar. If we drew a bar that started at a different value than 0, then the length of the bar and the bar endpoint would convey contradicting information. Such figures are internally inconsistent, because they show two different values with the same graphical element. Contrast this to a scenario where we visualize the data value with a dot. In this case, the value is only encoded in the location of the dot, not in the size or shape of the dot.

Similar issues will arise whenever we use graphical elements such as bars, rectangles, shaded areas of arbitrary shape, or any other elements that have a defined visual extent which can be either consistent or inconsistent with the data value shown. In all these cases, we need to make sure that there is no inconsistency. This concept has been termed as the *principle of proportional ink* [Bergstrom and West 2016]:

> When a shaded region is used to represent a numerical value, the area of that shaded region should be directly proportional to the corresponding value.

(It is common practice to use the word "ink" to refer to any part of a visualization that deviates from the background color. This includes lines, points, shared areas, and text. In this chapter, however, we are talking primarily about shaded areas.) Violations of this principle are quite common, in particular in the popular press and in the world of finance.

Visualizations Along Linear Axes

We first consider the most common scenario, visualization of amounts along a linear scale. Figure 17-1 shows the median income in the five counties that make up the state of Hawaii. It is a typical figure one might encounter in a newspaper article. A quick glance at the figure suggests that the county of Hawaii is incredibly poor while the county of Honolulu is much richer than the other counties. However, Figure 17-1 is quite misleading, because all the bars begin at $50,000 median income. Thus, while the endpoint of each bar correctly represents the actual median income in each county, the bar height represents the extent to which median incomes exceed $50,000, an arbitrary number. And human perception is such that the bar height is the key quantity we perceive when looking at this figure, not the location of the bar endpoint relative to the *y* axis.

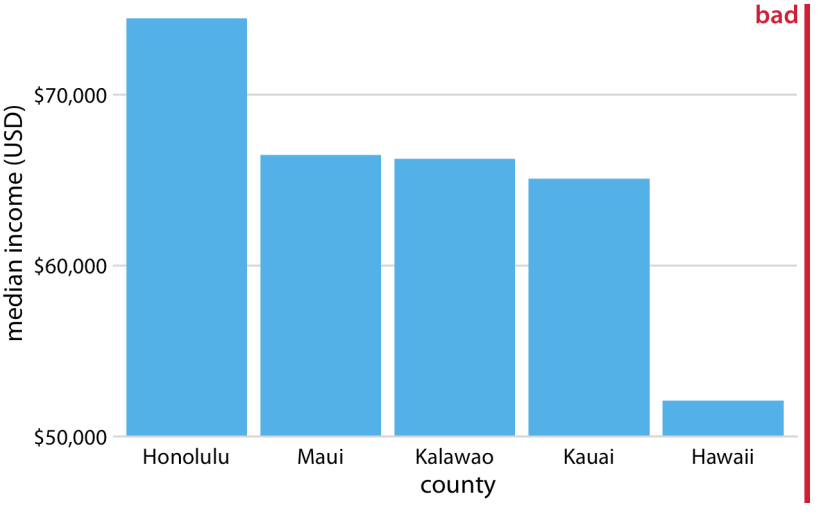


Figure 17-1. Median income in the five counties of the state of Hawaii. This figure is misleading, because the y*-axis scale starts at $50,000 instead of $0. As a result, the bar heights are not proportional to the values shown, and the income differential between the county of Hawaii and the other four counties appears much bigger than it actually is. Data source: 2015 Five-Year American Community Survey.*

An appropriate visualization of this dataset makes for a less exciting story (Figure 17-2). While there are differences in median income between the counties, they are nowhere near as big as Figure 17-1 suggested. Overall, the median incomes in the different counties are somewhat comparable.

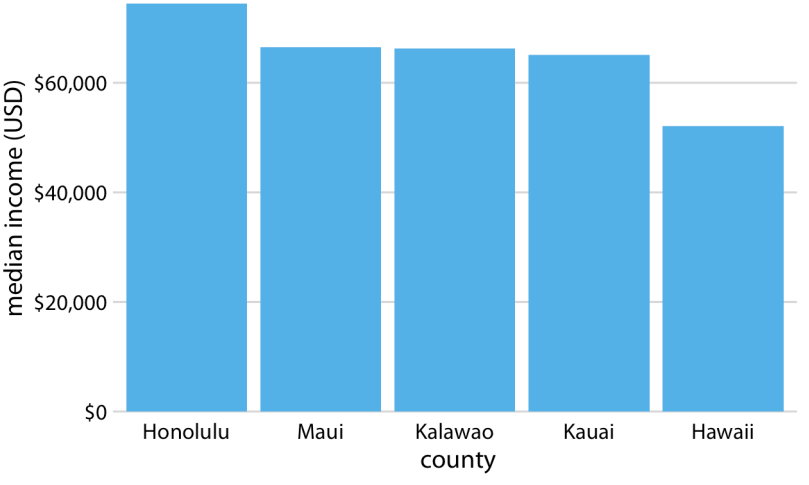


Figure 17-2. Median income in the five counties of the state of Hawaii. Here, the y*-axis scale starts at $0 and therefore the relative magnitudes of the median incomes in the five counties are accurately shown. Data source: 2015 Five-Year American Community Survey.*



Bars on a linear scale should always start at 0.

Similar visualization problems frequently arise in the visualization of time series, such as those of stock prices. Figure 17-3 suggests a massive collapse in the stock price of Facebook occurred around Nov. 1, 2016. In reality, the price decline was moderate relative to the total price of the stock (Figure 17-4). The *y*-axis range in Figure 17-3 would be questionable even without the shading underneath the curve. But with the shading, the figure becomes particularly problematic. The shading emphasizes the distance from the location of the *x* axis to the specific *y* values shown, and thus it creates the visual impression that the height of the shaded area at a given day represents the stock price of that day. Instead, it only represents the difference in the stock price from the baseline, which is $110 in Figure 17-3.

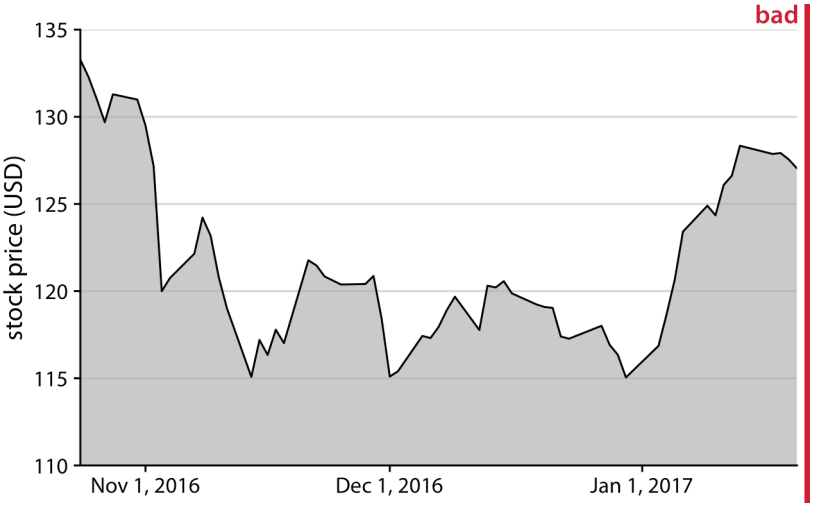


Figure 17-3. Stock price of Facebook (FB) from Oct. 22, 2016 to Jan. 21, 2017. This figure seems to imply that the FB stock price collapsed around Nov. 1, 2016. However, this is misleading, because the y *axis starts at $110 instead of $0. Data source: Yahoo! Finance.*

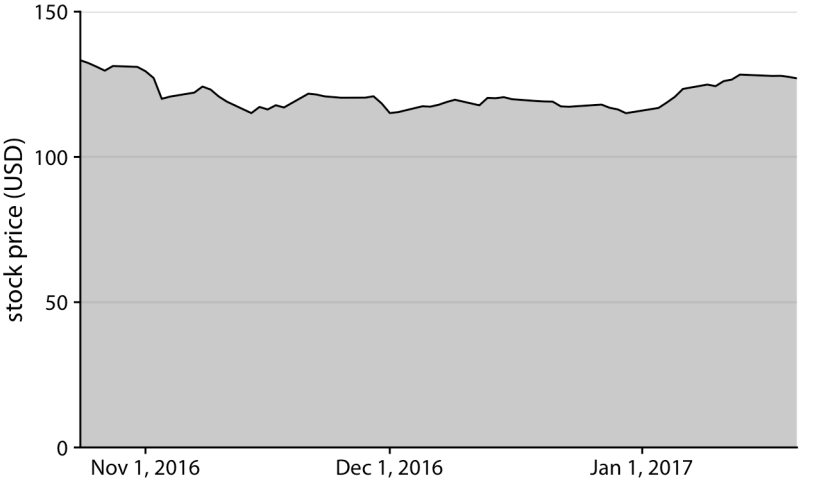


Figure 17-4. Stock price of Facebook (FB) from Oct. 22, 2016 to Jan. 21, 2017. By showing the stock price on a y *scale from $0 to $150, this figure more accurately relays the magnitude of the FB price drop around Nov. 1, 2016. Data source: Yahoo! Finance.*

The examples of Figures 17-2 and 17-4 could suggest that bars and shaded areas are not useful to represent small changes over time or differences between conditions, since we always have to draw the whole bar or area starting from 0. However, this is not the case. It is perfectly valid to use bars or shaded areas to show differences between conditions, as long as we make it explicit which differences we are showing. For example, we can use bars to visualize the change in median income in Hawaiian counties from 2010 to 2015 (Figure 17-5). For all counties except Kalawao, this change amounts to less than $5,000. (Kalawao is an unusual county, with fewer than 100 inhabitants, and it can experience large swings in median income from a small number of people moving into or out of the county.) And for Hawaii County, the change is negative; i.e., the median income in 2015 was lower than it was in 2010. We represent negative values by drawing bars that go in the opposite direction, extending downward from 0 rather than up.

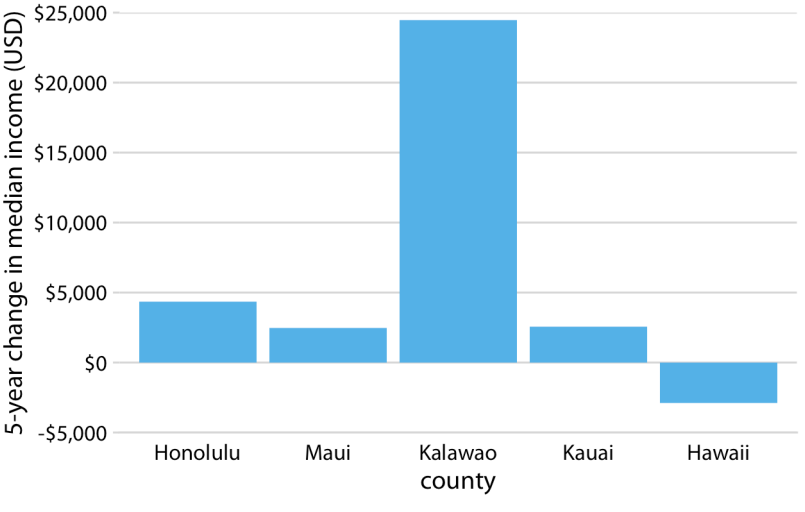


Figure 17-5. Change in median income in Hawaiian counties from 2010 to 2015. Data source: 2010 and 2015 Five-Year American Community Surveys.

Similarly, we can draw the change in Facebook stock price over time as the difference from its temporary high point on Oct. 22, 2016 (Figure 17-6). By shading an area that represents the distance from the high point, we are accurately representing the absolute magnitude of the price drop without making any implicit statement about the magnitude of the price drop relative to the total stock price.

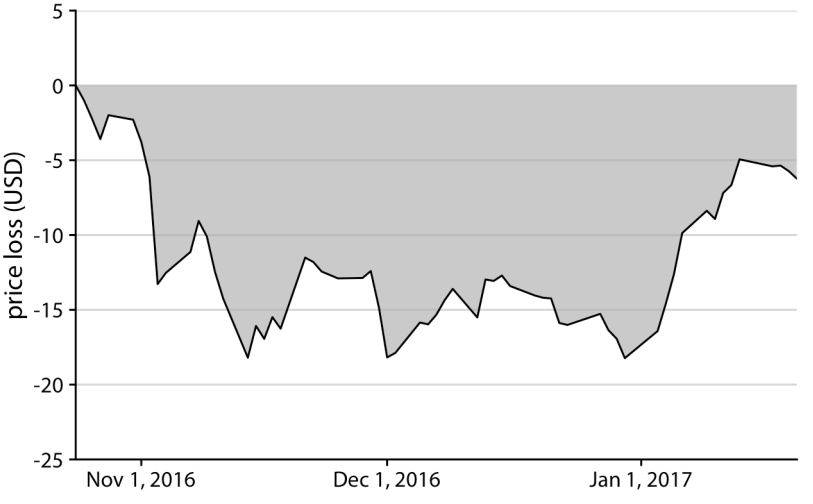


Figure 17-6. Decline in Facebook (FB) stock price relative to the price of Oct. 22, 2016. Between Nov. 1, 2016 and Jan. 1, 2017, the price remained approximately $15 lower than it was at its high point on Oct. 22, 2016. The price started to recover in January. Data source: Yahoo! Finance.

Visualizations Along Logarithmic Axes

When we are visualizing data along a linear scale, the areas of bars, rectangles, or other shapes are automatically proportional to the data values. The same is not true if we are using a logarithmic scale, because data values are not linearly spaced along the axis. Therefore, one could argue that, for example, bar graphs on a log scale are inherently flawed. On the flip side, the area of each bar will be proportional to the logarithm of the data value, and thus bar graphs on a log scale satisfy the principle of proportional ink in log-transformed coordinates. In practice, I think neither of these two arguments can resolve whether log-scale bar graphs are appropriate. Instead, the relevant question is whether we want to visualize amounts or ratios.

In Chapter 3, I have explained that a log scale is the natural scale to visualize ratios, because a unit step along a log scale corresponds to multiplication with or division by a constant factor. In practice, however, log scales are often used not specifically to visualize ratios but rather just because the numbers shown vary over many orders of magnitude. As an example, consider the gross domestic products (GDPs) of countries in Oceania. In 2007, these varied from less than a billion US dollars (USD) to over 300 billion USD (Figure 17-7). Visualizing these numbers on a linear scale would not work, because the two countries with the largest GDPs (New Zealand and Australia) would dominate the figure.

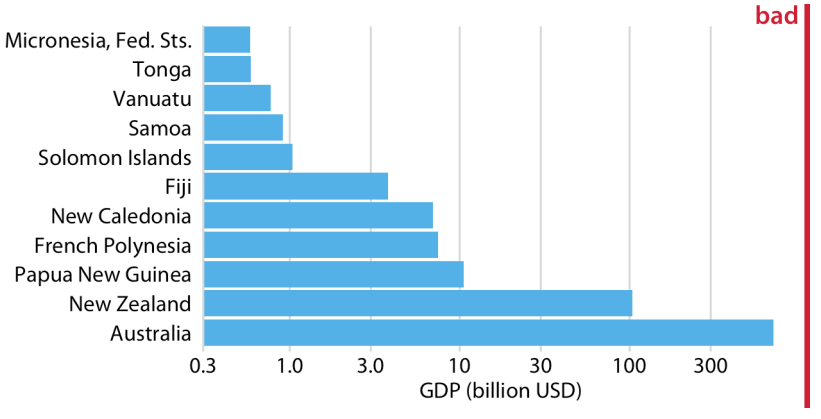


Figure 17-7. GDP in 2007 of countries in Oceania. The lengths of the bars do not accurately reflect the data values shown, since bars start at the arbitrary value of 0.3 billion USD. Data source: Gapminder.

However, the visualization with bars on a log scale (Figure 17-7) does not work either. The bars start at an arbitrary value of 0.3 billion USD, and at a minimum the figure suffers from the same problem of Figure 17-1, that the bar lengths are not representative of the data values. The added difficulty with a log scale, though, is that we cannot simply let the bars start at 0. In Figure 17-7, the value 0 would lie infinitely far to the left. Therefore, we could make our bars arbitrarily long by pushing their origin further and further away, as in Figure 17-8. This problem always arises when we try to visualize amounts (which is what the GDP values are) on a log scale.

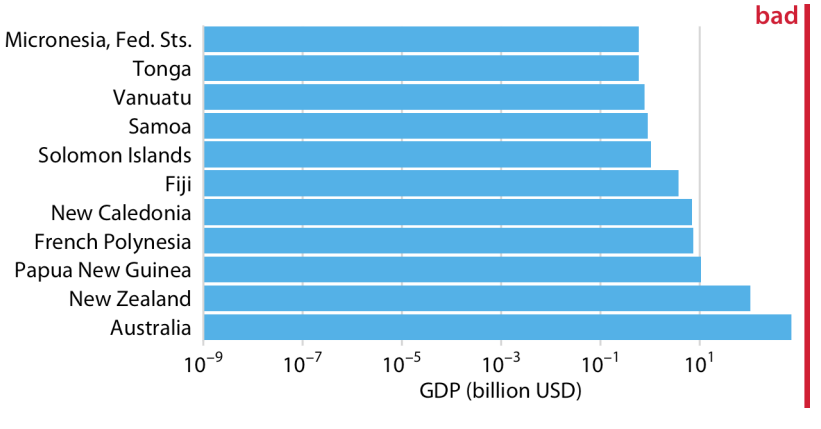


Figure 17-8. GDP in 2007 of countries in Oceania. The lengths of the bars do not accurately reflect the data values shown, since bars start at the arbitrary value of 10^{-9} billion USD. Data source: Gapminder.

For the data of Figure 17-7, I think bars are inappropriate. Instead, we can simply place a dot at the appropriate location along the scale for each country's GDP and avoid the issue of bar lengths altogether (Figure 17-9). Importantly, by placing the country names right next to the dots rather than along the *y* axis, we avoid generating the visual perception of a magnitude conveyed by the distance from the country name to the dot.

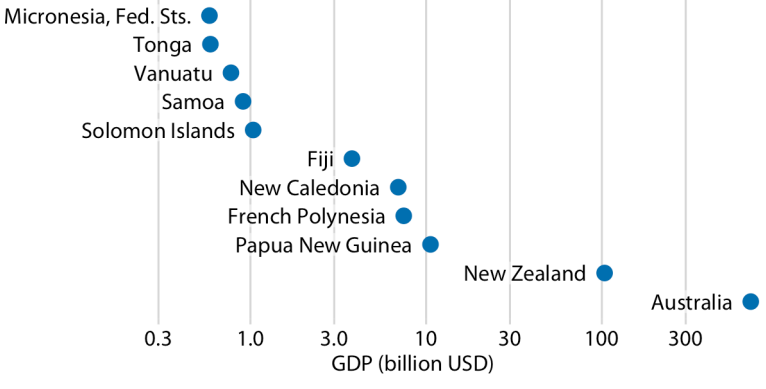


Figure 17-9. GDP in 2007 of countries in Oceania. Data source: Gapminder.

If we want to visualize ratios rather than amounts, however, bars on a log scale are a perfectly good option. In fact, they are preferable to bars on a linear scale in that case. As an example, let's visualize the GDP values of countries in Oceania relative to the GDP of Papua New Guinea. The resulting figure does a good job of highlighting the key relationships between the GDPs of the various countries (Figure 17-10). We can see that New Zealand has over 8 times the GDP of Papua New Guinea and Australia over 64 times, while Tonga and the Federated States of Micronesia each have less than 1/16th the GDP of Papua New Guinea. French Polynesia and New Caledonia are close but have slightly smaller GDPs than Papua New Guinea.

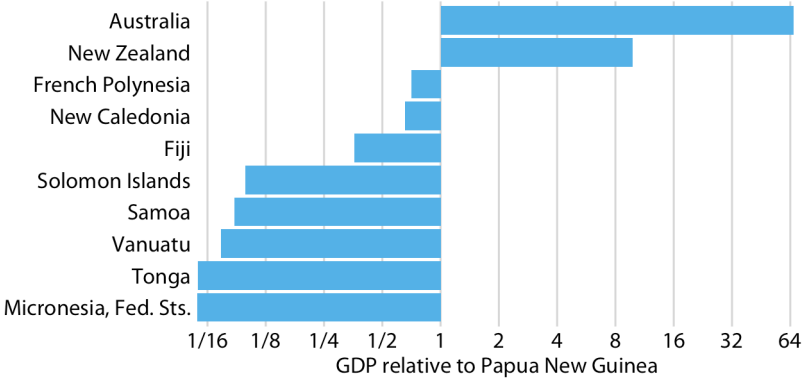


Figure 17-10. GDP in 2007 of countries in Oceania, relative to the GDP of Papua New Guinea. Data source: Gapminder.

Figure 17-10 also highlights that the natural midpoint of a log scale is 1, with bars representing numbers above 1 going in one direction and bars representing numbers below 1 going in the other direction. Bars on a log scale represent ratios and should always start at 1, and bars on a linear scale represent amounts and should always start at 0.



When bars are drawn on a log scale, they represent ratios and need to be drawn starting from 1, not 0.

Direct Area Visualizations

All the preceding examples visualized data along one linear dimension, so that each data value was encoded both by area and by location along the x or y axis. In these cases, we can consider the area encoding as incidental and secondary to the location encoding of the data value. Other visualization approaches, however, represent the data values primarily or directly by area, without a corresponding location mapping. The most common one is the pie chart (Figure 17-11). Even though technically the data values are mapped onto angles, which are represented by location along a circular axis, in practice we are typically not judging the angles of a pie chart. Instead, the dominant visual property we notice is the area of each pie wedge.

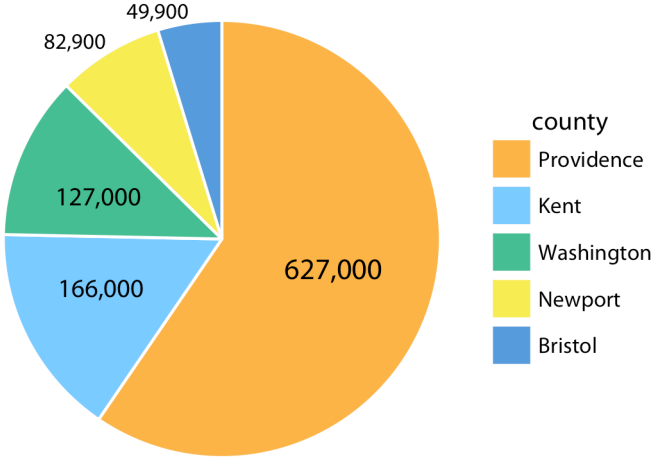


Figure 17-11. Number of inhabitants in Rhode Island counties, shown as a pie chart. Both the angle and the area of each pie wedge are proportional to the number of inhabitants in the respective county. Data source: 2010 US Decennial Census.

Because the area of each pie wedge is proportional to its angle, which is proportional to the data value the wedge represents, pie charts satisfy the principle of proportional ink. However, we perceive the area in a pie chart differently from the same area in a bar plot. The fundamental reason is that human perception primarily judges distances and not areas. Thus, if a data value is encoded entirely as a distance, as is the case with the length of a bar, we perceive it more accurately than when the data value is encoded through a combination of two or more distances that jointly create an area. To see this difference, compare Figure 17-11 to Figure 17-12, which shows the same data as bars. The difference in the number of inhabitants between Providence County and the other counties appears larger in Figure 17-12 than in Figure 17-11.

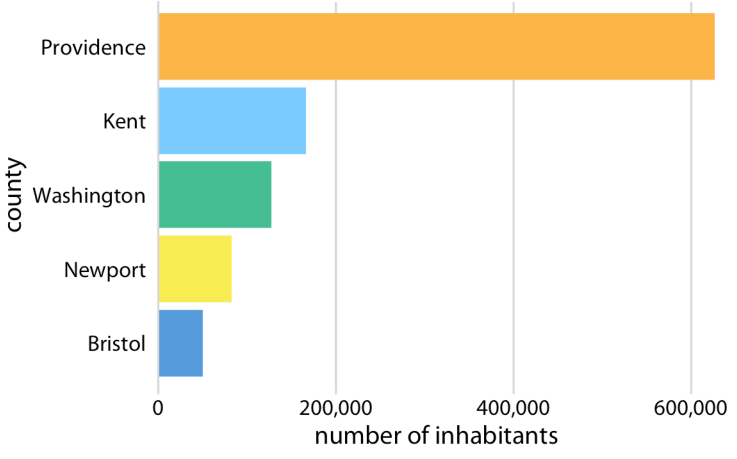


Figure 17-12. Number of inhabitants in Rhode Island counties, shown as bars. The length of each bar is proportional to the number of inhabitants in the respective county. Data source: 2010 US Decennial Census.

The problem that human perception is better at judging distances than at judging areas also arises in treemaps (Figure 17-13), which can be thought of as square versions of pie charts. Again, in comparison to Figure 17-12, the differences in the number of inhabitants among the counties appears less pronounced in Figure 17-13.

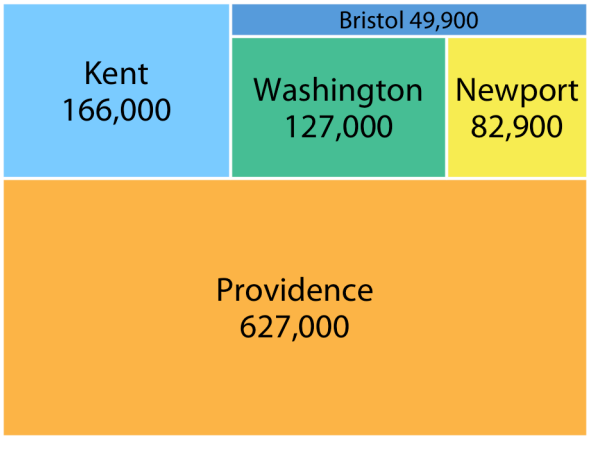


Figure 17-13. Number of inhabitants in Rhode Island counties, shown as a treemap. The area of each rectangle is proportional to the number of inhabitants in the respective county. Data source: 2010 US Decennial Census.